

Toxicity Classification in YouTube Comments

Deep Learning Final Project Defense

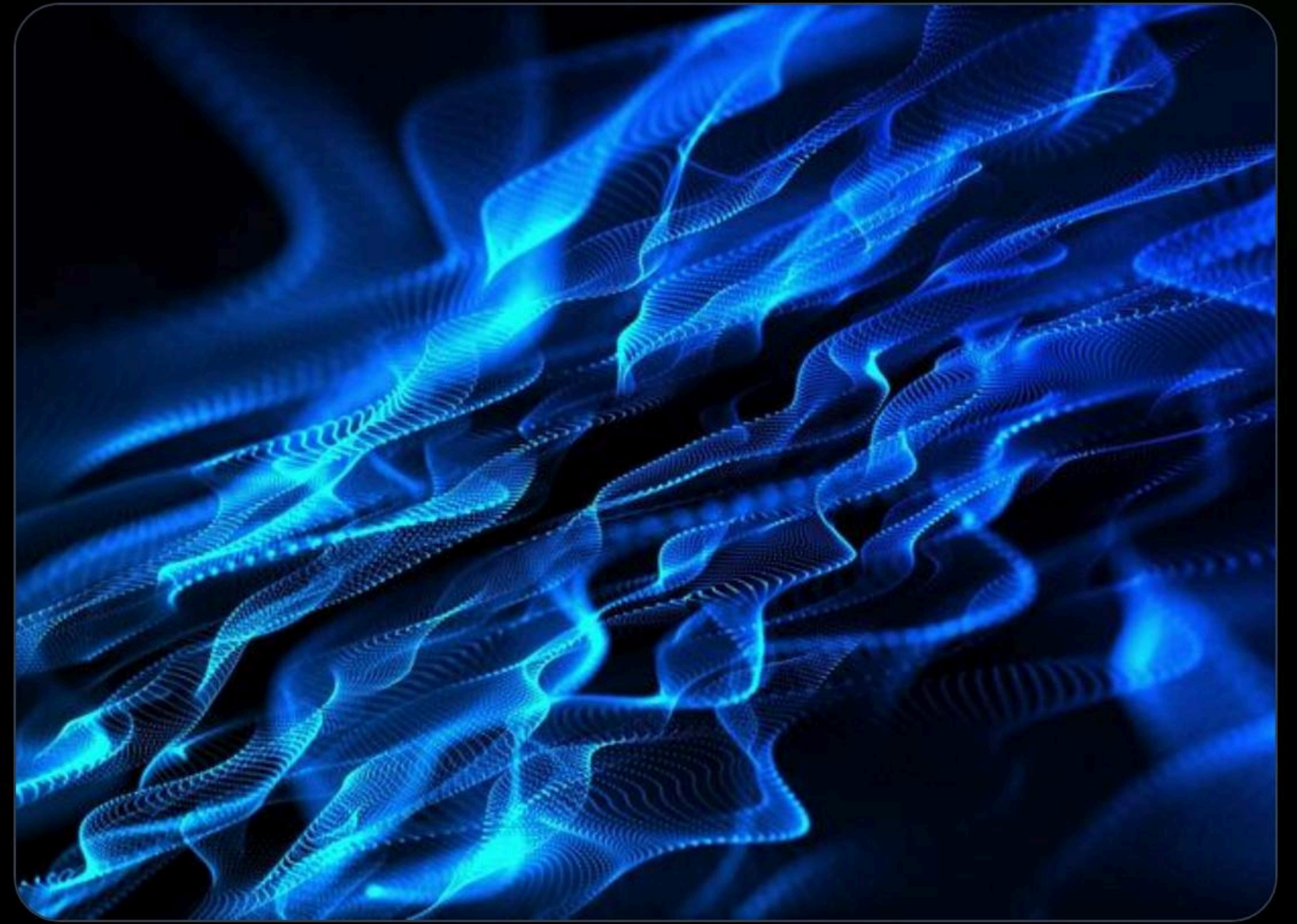
Bauyrzhan Turebek • Applied Deep Learning

| Problem Statement

The Scale: YouTube generates billions of comments daily. Manual moderation is technically impossible.

The Impact: Toxic behavior, threats, and hate speech drive away users and creators, damaging platform health.

The Solution: An automated **Multi-label classification system** capable of understanding nuanced human language.



The Dataset: Jigsaw Toxic Data



159K Samples

Extracted from Wikipedia talk pages
(similar to YouTube comment structure).



6 Labels

Toxic, Severe Toxic, Obscene, Threat,
Insult, and Identity Hate.

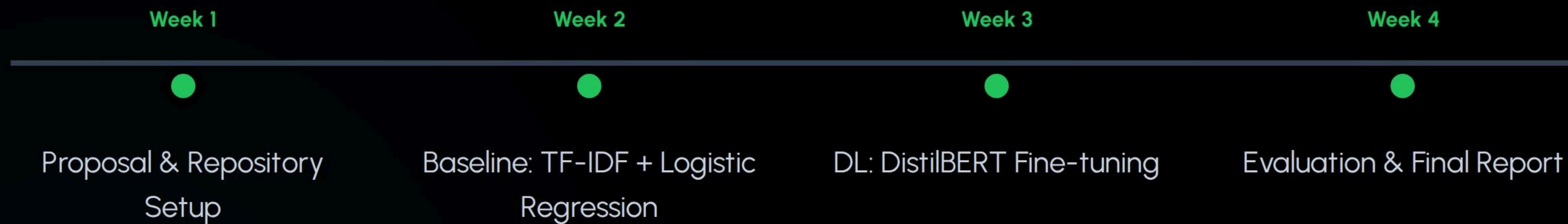


Imbalance

Severe class imbalance. Minority
classes (Threat) represent < 1% of data.

*Note: To handle memory constraints, we performed experiments on a **10% stratified sample**.*

Project Roadmap



| Baseline: Statistical Approach

Logistic Regression + TF-IDF

- **Vectorization:** 10,000 most frequent n-grams.
- **Architecture:** Binary Relevance (One-vs-Rest).
- **Pros:** Extremely fast, highly interpretable.
- **Cons:** "Bag-of-words" misses word order and semantic context.

Baseline Result

~ 0.65 F1-Score

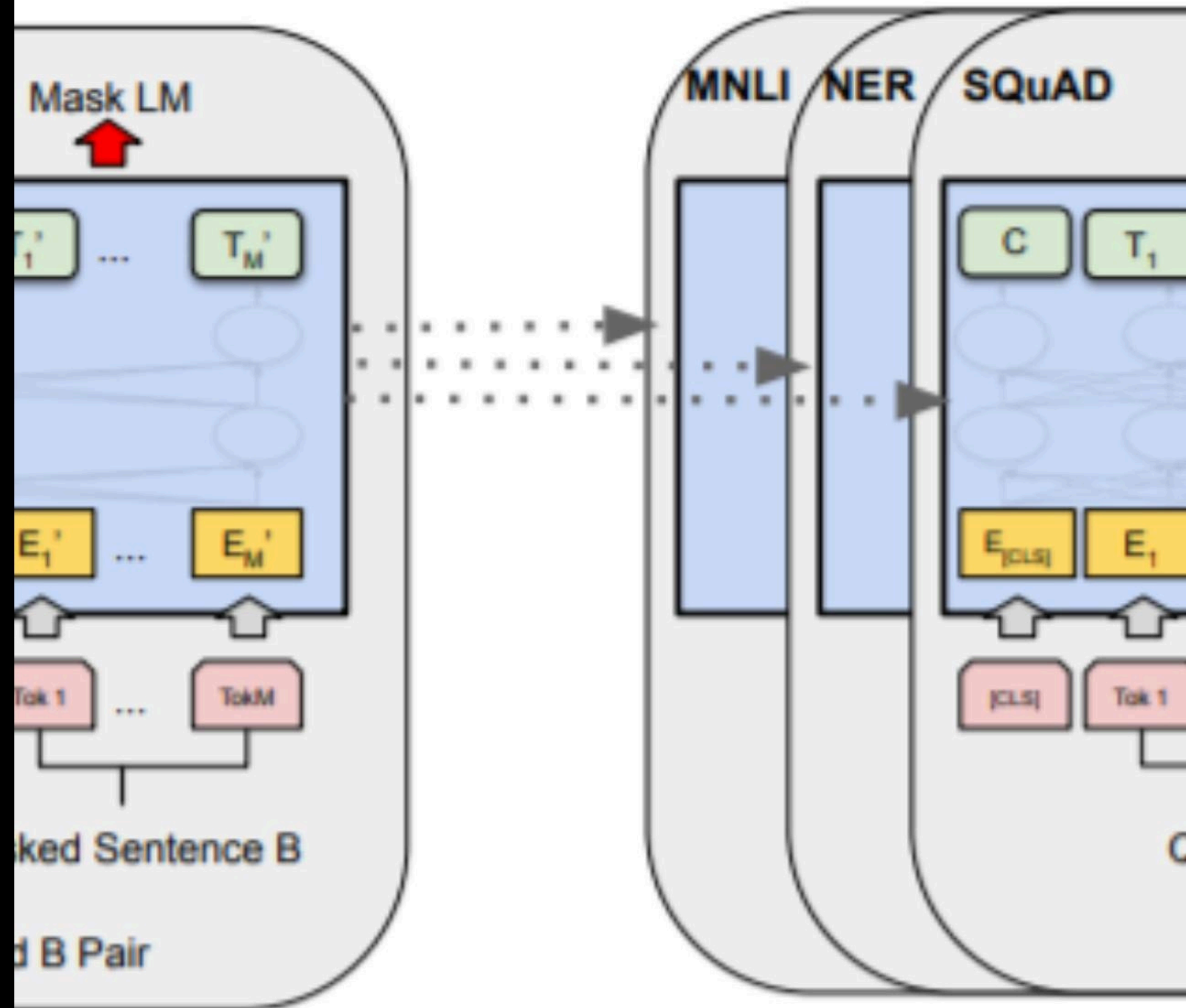
Decent for keywords, poor for sarcasm.

DistilBERT Fine-Tuning

Architecture: A distilled version of BERT, retaining 97% of its performance while being 40% smaller and 60% faster.

Method: Knowledge Distillation (Teacher-Student network).

- **Hardware:** Google Colab T4 GPU
- **Loss:** BCEWithLogitsLoss
- **Optimizer:** AdamW



Final Results: DistilBERT

Category	Precision	Recall	F1-Score
Toxic	0.91	0.79	0.85
Obscene	0.81	0.89	0.84
Insult	0.75	0.79	0.77
Severe Toxic	0.82	0.07	0.14
Identity Hate	0.00	0.00	0.00

Micro Average F1-Score: 0.78

| Critical Success & Failure

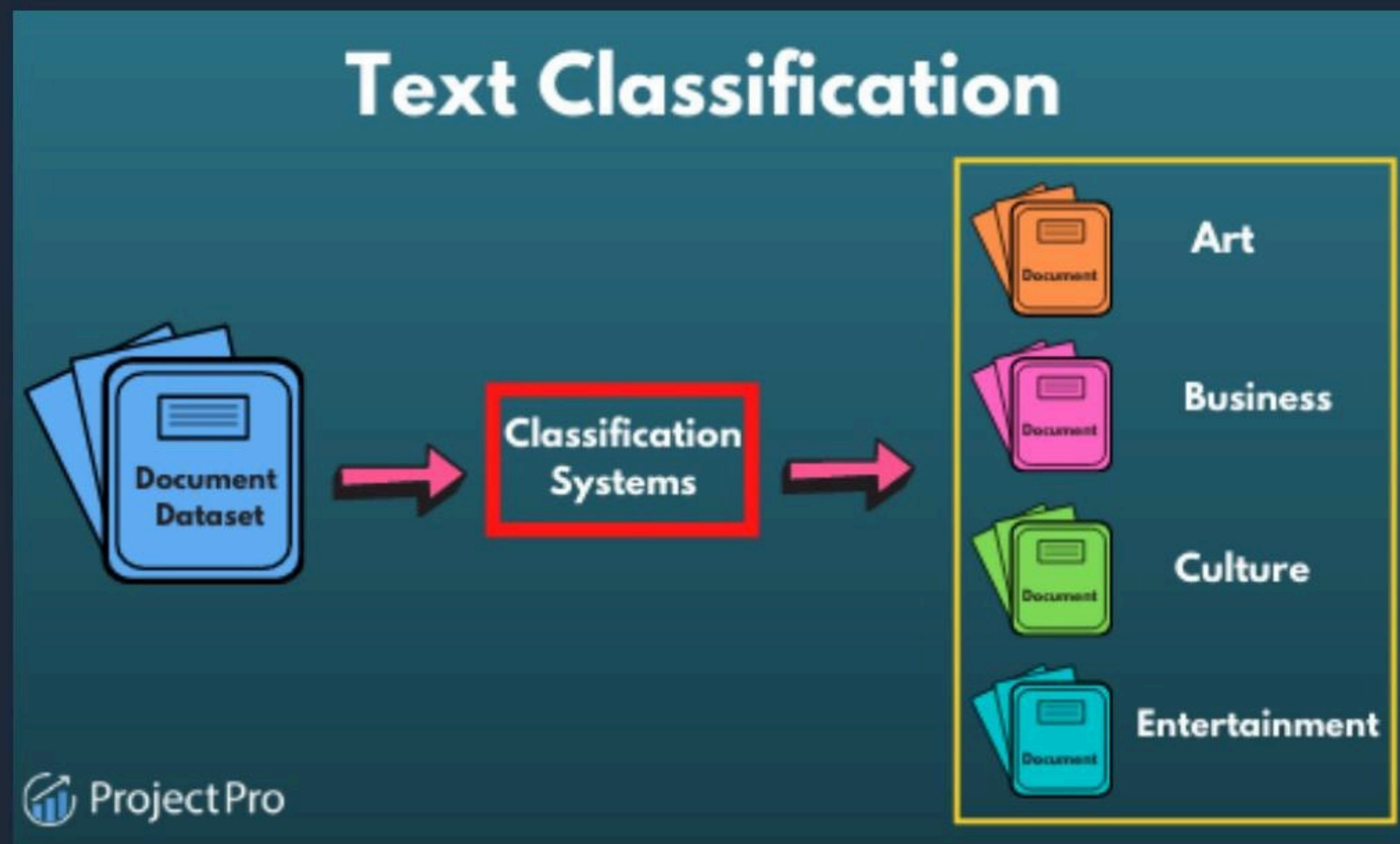
✓ Success Case

Excellent detection in **Toxic** and **Obscene** classes. DistilBERT captures the harmful sentiment even with complex sentence structures.

! Failure Case

Threat (0.00 F1): Due to using a 10% sample, rare classes had too few training instances (< 30 examples) for the model to generalize.

Internal Representations



Contextual Embeddings

Unlike TF-IDF, DistilBERT "understands" that the same word can be toxic or benign depending on the surrounding tokens.

Fine-tuning: We updated only the final 6 layers and the classification head to adapt the pre-trained knowledge to the YouTube domain.

| Limitations & Future

Implicit Toxicity

Sarcasm and coded hate speech are still challenging for base DistilBERT.

Class Imbalance

Future work: Use Oversampling or Data Augmentation for rare labels.

Multi-lingual

Current model is English-only. Plan: Switch to XLM-RoBERTa for local markets.

Project Complete

End-to-end pipeline deployed on GitHub.

Key Metric: +13% F1-score improvement over baseline

Thank You! Q&A

Ready to answer questions from the exam papers.

